

DataCraft

Consumer Behavior & Shopping Analytics — Business Problem

1. Business Context

A leading retail company wants to gain a clearer understanding of how its customers shop and what influences their purchasing decisions. Management has observed changes in buying patterns across customer demographics, product categories, and sales channels, including online and offline purchases. The business aims to use these patterns to strengthen sales performance, improve customer satisfaction, and build long-term customer loyalty.

2. Business Challenge

The company needs to identify the factors that are most closely associated with consumer purchasing behavior and repeat purchases. Particular attention is given to the role of discounts, customer reviews, seasonal effects, payment preferences, and differences between shopping channels.

3. Key Business Question

How can consumer shopping data be used to identify important purchasing trends, strengthen customer engagement, and improve marketing and product strategies?

4. Project Objectives

- Prepare and transform the raw consumer dataset into a reliable format suitable for analysis.
- Examine customer segments, purchasing behavior, loyalty patterns, and potential purchase drivers.
- Use structured analysis to uncover relationships and trends within the consumer data.
- Create an interactive Power BI dashboard that communicates important patterns and findings clearly.
- Translate analytical findings into practical recommendations for marketing, product, and customer-engagement decisions.

5. Analytical Approach

Python: Data cleaning, transformation, preprocessing, and exploratory analysis.

SQL: Structured data analysis, transaction-oriented queries, customer segmentation, loyalty analysis, and purchase-driver investigation.

Power BI: Interactive dashboards and visual analysis of key trends and patterns.

Reporting: Consolidation of findings into business insights and actionable recommendations.

6. Expected Business Value

The analysis is intended to help stakeholders understand customer behavior more effectively and identify opportunities to improve engagement, encourage repeat purchases, refine marketing strategies, and make better-informed product decisions.

7. Deliverables

- Cleaned and analysis-ready dataset.
- Python analysis and preprocessing files.
- SQL queries and analytical outputs.
- Interactive Power BI dashboard.
- Project report containing key findings and recommendations.
- Presentation summarizing insights and proposed actions.

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